SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct

I S A Balaji (192126011) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1	1	1
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

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21/18

1. Missing OSI layer and explain their function

The missing layers are session layer and Data link layer.

7. Application.
Provides network services to user

↓

6. Presentation
Data translation, encryption and compression

↓

5. Session
establishes and manages communication

↓

4. transport.
Reliable end to end data delivery.

↓

3. Network.
Routing and logically addressing.

↓

2. Data link.
Framing, MAC addressing and error detection

↓

Total Marks 17.00

1. Physical

Transmit raw bits through the medium.

Importance for reliable communication:

-> The session layer manages and maintain communication between devices.

-> The Data link layer transfers frames b/w directly connected devices and detects transmission error.

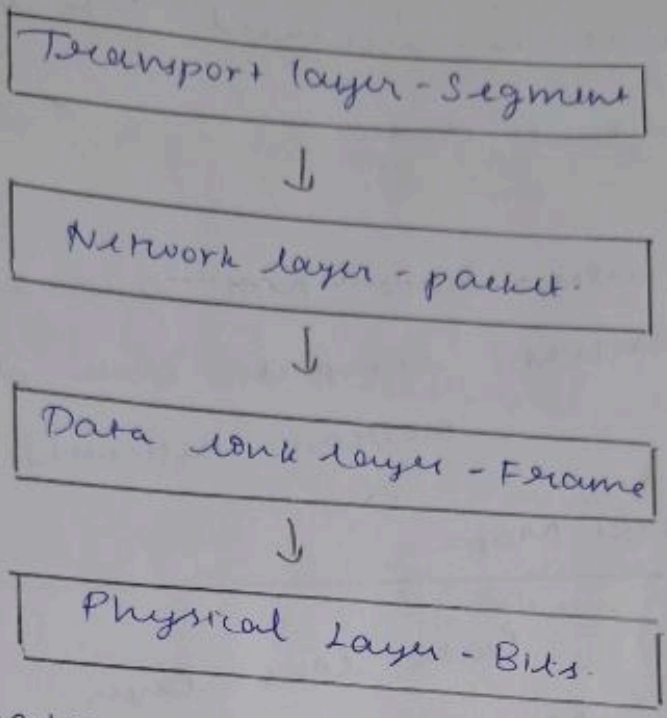
-> Together with the other OSI layers, they support organized and reliable data communication.

2- Map data units to other OSI layer and explain data transformation.

OSI layer	Data Unit
Transport	segment
Network	Packet
Data link	frame
Physical	Bits

Total Marks 11 /30

Concept Map:



Explanation:

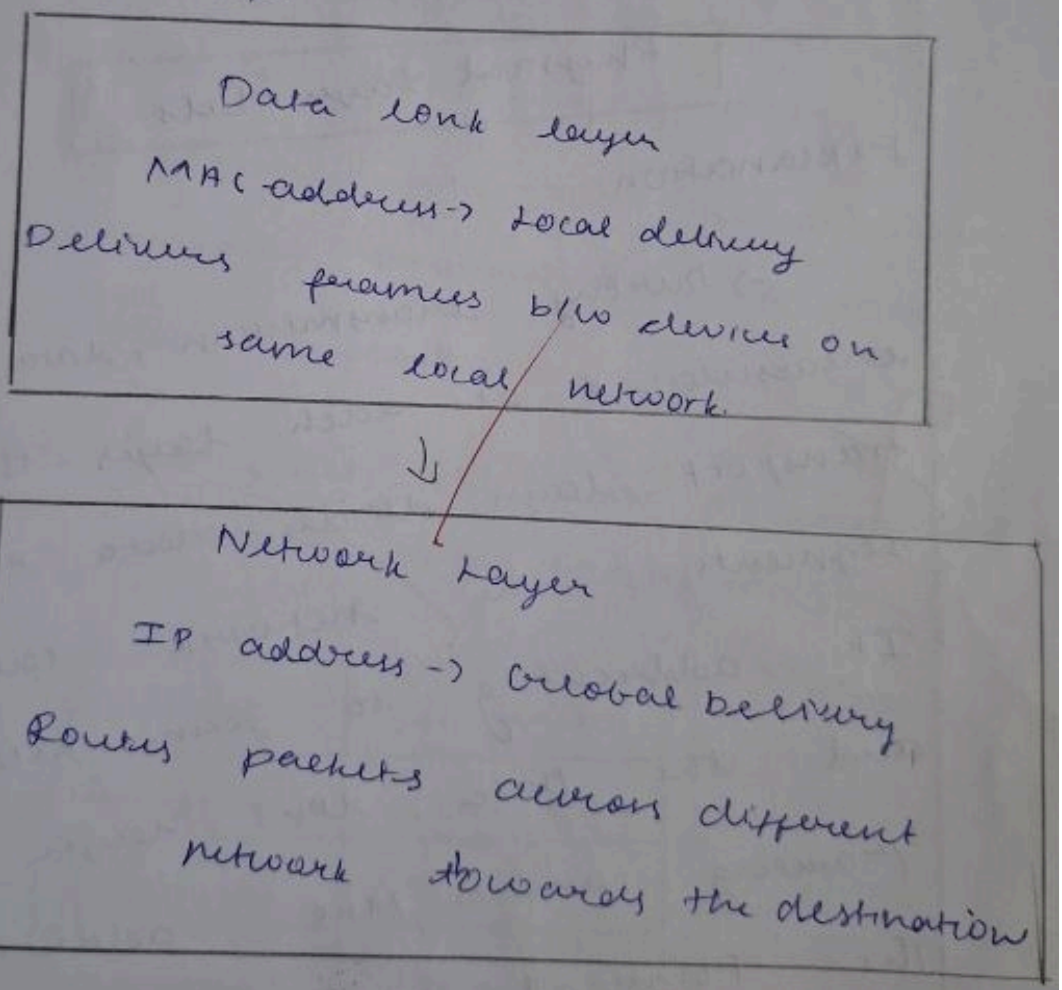
→ During transmission, data is encapsulated at each layer. The transport layer divides data into segments, the network layer adds IP addressing to form packets and the Data link layer adds framing and MAC addressing. The Physical layer transmits the information as bits.

Total Marks 11 /30

-> At the receiving end, the process is reversed to recover the original data.

3. Explain how MAC and IP addresses ensure complete data delivery across networks.

Concept Map:



Total Marks 11 /20

Explanation:

-> MAC address: Identifies a network interface for delivery within a local network.

-> IP address: Identifies the source and destination at the network layer and helps routing forward packets across networks.

-> Combined operation: IP addressing supports end to end routing, while MAC addressing supports delivery over each local link.

* Thus, both addressing system work together to deliver data to the correct destination across networks.

Total Marks 18 /30